

Classification of Gout from Chief Complaints in the Emergency Department: Comparison of Feature-based and Representation Learning Approaches

LHS 712 Winter 2026

Medha Rao

Abstract

Gout is the most common inflammatory arthritis in the United States and results in 200,000 emergency department (ED) visits annually. Due to the timing of ICD billing assignment, gout patients cannot be prospectively identified during triage. This leads to missed referrals and perpetuates the gap in care for gout patients. Chief complaints, which are recorded at triage time, are a real-time data source for prospective gout identification using natural language processing (NLP). This study compares five NLP methods: support vector machine (SVM) and random forest (RF) with TF-IDF features, fine-tuned BERT, ClinicalBERT, and BioMED RoBERTa. The task was to classify gout from ED chief complaints, and propose an optimal classification threshold to be used for potential clinical alert systems. Data came from the PhysioNet corpus of 8,437 ED chief complaints, of which 112 (1.3%) were gout cases. This resulted in a 74.3:1 class imbalance. Representation learning methods overall outperformed the feature-based methods, with ClinicalBERT achieving the best test f1 of 0.79, test macro f1 of 0.80, test precision of 0.81, and test recall of 0.76. Threshold analysis on ClinicalBERT identified 0.60 as the optimal threshold value. This suggests that ClinicalBERT deployed with a 0.60 threshold could serve as a viable real-time clinical alert system for gout identification during ED triage, however external validation is needed before clinical deployment.

Introduction

Gout is the most common inflammatory arthritis in the United States and affects over 9 million patients.¹ Although there is a prevalence of gout in the US, it remains undertreated and understudied, especially in the emergency department (ED).¹ Approximately 200,000 ED visits annually are due to gout, where acute flares are managed but outpatient care rarely follows.² Referrals to rheumatologists or uric acid lowering therapies are frequently missed, thus perpetuating this gap in long-term care.²

One reason for this gap is the timing of ICD billing code assignments.^{2,3} These codes are typically assigned after the ED visit is complete, which means they cannot be used to prospectively identify gout patients during ED triage time. Chief complaints, which are recorded by nursing staff at ED triage time, are one of the earliest documented sources of clinical information in an ED visit and are available in real-time.^{3,4} Natural language processing (NLP) applied to this corpus is a promising method for prospective gout identification.

Given the large, growing landscape of NLP techniques, ranging from traditional feature-based methods such as support vector machines and random forests with TF-IDF features, to more modern representation learning methods using fine-tuned transformer models, there is no consensus about the best model approach for identifying gout flares from ED chief

complaints. Additionally, making a deployable clinical alert system would require careful consideration of classification thresholds and the tradeoff between precision and recall in the ED setting.

This study compares feature-based and representation learning NLP approaches for the classification of gout from ED chief complaints, identifies the best method, and proposes an optimal threshold for gout classification to be used in a real-time clinical alert system.

Methods

Data and Study Population

Data came from PhysioNet, where a corpus of nurse triage chief complaints from an ED at an academic medical center in the deep south were collected over the span of a month in 2019 and 2020.⁵ The 2019 set had 300 chief complaints collected, and the 2020 set had 8,137. Data was de-identified according to HIPAA guidelines. The study population was 54% female, and 55% Black, 40% White, and 3% Hispanic/Latino or Asian.⁵

The data consists of a “Chief Complaint,” “Predict,” and “Consensus” column (**Table 1**). The “Predict” column was constructed by expert review by a rheumatologist (MD) and an informatician (PhD) who annotated the chief complaints as indicative of gout (Y), not indicative of gout (N), or unknown (U). Any observations not labelled received an unmarked label (-). The “Consensus” column was constructed by expert review by a separate rheumatologist (MD) and post-doctoral fellow (GRA) who annotated the chief complaints after consulting the patient’s chart as well.

Table 1. Four selected observations from the 2020 redacted data set.

Chief Complaint	Predict	Consensus
"I dont know whats going on with my head, its not a regular HA, I passed out when I was home on the bed" also c/o of ear pain and right eye pain. PMHX PMHX HA,	N	-
"yesterday around 0917 am my R knee just popped. I have pain in my knee and it goes up to my thigh." Pt ambulates with walker. Denies numbness/tingling. Pmhx: gout, OA.	U	N
L knee pain for 2 weeks, unable to walk for 2 days . Non compliant with gout meds. Lknee edematous without erythema. PMHx; gout HTN,	Y	Y
left wrist and hand swelling starting 2 weeks ago, c/o throbbing pn in hand, denies PMH	U	Y

Models Chosen

Five total models were chosen: support vector machine (SVM) with TF-IDF features (feature-based), random forest (RF) with TF-IDF features (feature-based), fine-tuned BERT (representation learning), fine-tuned ClinicalBERT (representation learning), and fine-tuned

BioMED RoBERTa (representation learning). These models were chosen to compare traditional feature-based approaches against domain-pretrained representation learning models.

Data Pre-Processing

The chief complaints were first lowercased, and all punctuation was removed. Whitespace was normalized, where leading and trailing whitespaces were removed, and only one space was preserved between each word. This was done across all five models built. For the two feature-based methods and the fine-tuned BERT model, an additional pre-processing step of regex medical abbreviation expansion was implemented. Medical abbreviations to expand were chosen based on the most common abbreviations found amongst the chief medical complaints, and common medical abbreviations identified in literature.^{6,7} ClinicalBERT and BioMED RoBERTa did not have medical abbreviation expansion applied, as these were trained using medical text such as clinical notes, biomedical text, or journal articles.

For the two feature-based models, TF-IDF vectorization using sublinear scaling, 1-, 2-grams and SMOTE oversampling was applied, with maximum features of 15,000. SMOTE oversampling was done to mitigate the large class imbalance between gout and non-gout cases observed in the data. For the three representation learning models, subword tokenization was enabled with a maximum sequence length of 128 tokens. This was done to handle potential rare or out-of-vocabulary instances while preventing the feature space from becoming too large.

Labels were taken from the “Consensus” column, and when the “Consensus” column was labelled as either “U” or “-,” the “Predict” column was used as a fallback. If *both* the “Consensus” and “Predict” columns had “U” or “-” the observation was treated as non-gout, reflecting real-world practice.

Finally, the dataset was split 70% training, 15% validation, and 15% test. This was a stratified split by gout or non-gout to ensure that there were not large discrepancies in the number of gout cases between the splits.

Model Configurations and Training

The SVM with TF-IDF features used 5-fold cross validation to find the optimal parameter, C , that maximized the f1 score. The RF with TF-IDF features also used 5-fold cross validation with random search ($n_{\text{iter}} = 20$) to tune the number of trees, maximum tree depth, and minimum samples required to split a node. The combination that maximized the f1 score was selected.

The three representation learning models used the AdamW optimizer with weight decay of 0.01, learning rate of $1e-5$, linear warmup scheduler over the first 10% of training, batch size of 32, weighted loss of 10:1 (gout:non-gout), and trained over 5 epochs. The models aimed to minimize the weighted cross-entropy loss, with the best validation f1 over the 5 epochs chosen as the final model. The weighted loss was a cap on the true ratio (74.3:1) to prevent training instability and ensure model convergence. To further prevent gradients from becoming too large, gradient clipping was applied with a max norm of 1.0.

Model Evaluation

Metrics used to evaluate each of the five models were: precision (gout), recall (gout), f1, and macro-f1. The best macro-f1 model was chosen for threshold analysis, as this equally weighted the f1 of both classes (gout and non-gout).

Threshold Analysis

The selected model underwent a threshold analysis, where a grid of 100 threshold values was tested, ranging from 0.05 to 0.95. The threshold that maximized the f1 score was selected.

Computing

The pipeline was built in a Jupyter notebook using Python. All computing was done through Google Colab, using the T4 GPU available. The random seed was set to 107 for reproducibility.

Results

The final dataset, combining both 2019 and 2020 data, had 8,437 observations. 112 (1.3%) of those observations were gout cases, leading to a 74.3:1 ratio for gout:non-gout. The train split had 5,905 observations with 78 (1.3%) gout cases, and the validation and test splits both had 1,266 observations each with 17 (1.3%) gout cases each. The chief complaints had a maximum character count of 266.0 for non-gout and 235.0 for gout, indicating that the chief complaints were not textually very burdensome. The mean text length for non-gout was 108.3 characters (sd = 52.5), and 94.3 characters (sd = 41.0) for gout cases (**Figure 1**).

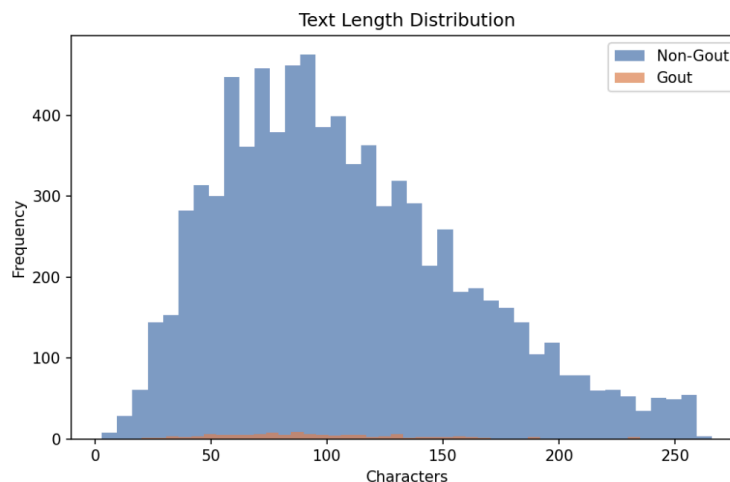


Figure 1. Text Length Distribution in full data set (n = 8,437)

The top 20 words were identified for the gout cases from the full data set before splits, of which “gout,” “pain,” “foot,” “swelling,” “knee,” and “flare” were identified as some of the top 20 words (see **Appendix A**). These top words for gout cases indicate that the chief complaints

carry enough information for gout cases specifically, and point to a potential self-diagnosis that some patients may be bringing into the ED.

After training each of the five models, precision (gout), recall (gout), f1, and macro-f1 were calculated on the validation set (**Table 2**). Recall was particularly low for the feature-based methods in the validation set. The RF had a recall of 0.06 but perfect precision, suggesting severe overfitting to SMOTE’s samples in the high-dimensional sparse TF-IDF feature space. The RF model learned to classify the minority samples well, but could not generalize those learnings to real gout cases. The SVM was considerably better than the RF, with a recall of 0.35 and macro-f1 of 0.75. These are still poorer metrics than the representation learning models. Amongst the representation learning models, BERT achieved the highest recall of 0.82 (amongst all models as well). However, ClinicalBERT and BioMED RoBERTa achieved a better balance of precision and recall, with BioMED RoBERTa’s highest validation f1 (0.76) and validation macro-f1 (0.88). As expected, both of the domain-pre-trained models outperformed the more general BERT fine-tuned model, supporting the hypothesis that the domain pretraining with clinical and biomedical domains respectively are at an advantage for this task.

Table 2. Validation set metrics

Metric	SVM	RF	BERT	ClinicalBERT	BioMED RoBERTa
Precision (gout)	0.86	1.00	0.61	0.71	0.92
Recall (gout)	0.35	0.06	0.82	0.71	0.65
f1	0.50	0.11	0.70	0.71	0.76
macro-f1	0.75	0.55	0.85	0.85	0.88

The test set results (**Table 3**) show that ClinicalBERT was the best overall model with the highest f1 (0.78), macro f1 (0.89), and recall (0.82). BERT and BioMED RoBERTa performed comparably, with f1 scores of 0.73 and 0.72 respectively. BERT’s recall matched ClinicalBERT’s (0.82) but at the cost of lower precision (0.64). BioMED RoBERTa saw a drop in recall from validation to test. ClinicalBERT generalized well from validation to test sets. The SVM showed improvement from validation to test sets, with the recall jumping from 0.35 to 0.47. The feature-based methods still underperform compared to the representation learning methods. Based on these results, ClinicalBERT was chosen as the best model and chosen for threshold analysis.

Table 3. Test set metrics

Metric	SVM	RF	BERT	ClinicalBERT	BioMED RoBERTa
---------------	------------	-----------	-------------	---------------------	-----------------------

Precision (gout)	1.00	1.00	0.59	0.81	0.72
Recall (gout)	0.47	0.06	0.76	0.76	0.76
f1	0.64	0.11	0.67	0.79	0.75
macro-f1	0.82	0.55	0.83	0.89	0.87

The threshold analysis revealed that the default threshold (0.50) was not far off from the optimal threshold of 0.60 (**Figure 2**). Precision, recall, and consequently f1, all remained relatively stable across thresholds. This suggests that the ClinicalBERT model outputs were fairly confident and not close to these threshold values. For a clinical alert system, the optimal threshold of 0.60 suggests that a slightly more conservative cutoff is appropriate to trade a bit of recall for improved precision and less false alarms.

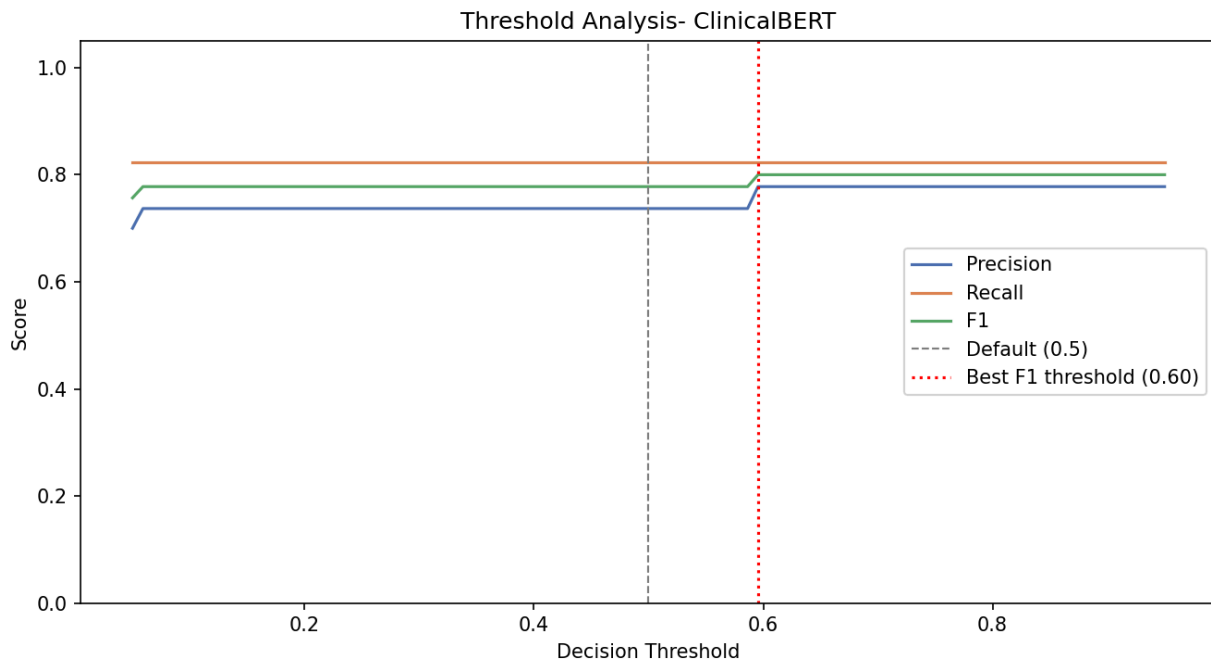


Figure 2. Threshold analysis for ClinicalBERT on test set.

Discussion

Overall, representation learning methods generally outperformed feature-based methods for gout identification from ED chief complaints. ClinicalBERT was the best model overall, achieving a test macro-f1 of 0.89, test precision of 0.81, and test recall of 0.76.

The worst model, RF, with high test precision (1.00) but extremely low test recall (0.06) was likely due to SMOTE sampling from 78 gout cases to try to balance the training set classes. SMOTE was not able to function in this high-dimensional sparse feature space. SVM had a test macro-f1 of 0.82, which is similar to the representation learning methods. While SVM is a

simpler model, it may be a reasonable alternative in a setting where computational resources are limited. Another comparison to note is ClinicalBERT outperforming BERT, which supports the hypothesis that domain pretraining is a factor in clinical NLP tasks. Additionally, ClinicalBERT outperforming BioMED RoBERTa was expected, because although both are domain specific, ClinicalBERT's pretraining corpus would more closely resemble the style of ED chief complaints.

A real-time alert system using ClinicalBERT with the identified optimal threshold of 0.60 could flag potential gout cases during triage, which would enable timely rheumatology referrals and uric acid lowering therapies from the ED, as opposed to the standard flow of care in the ED which would be acute pain management. The tradeoff between precision and recall is clinically meaningful, as low precision (false alarms) would unnecessarily burden clinical staff and low recall (missed cases) would perpetuate the gap in care for gout patients. ClinicalBERT achieves a balance of precision and recall (0.81 and 0.76 respectively), supporting the results of ClinicalBERT being the best model for this task.

While this study was able to identify a superior model and suggest an optimal threshold for clinical alert systems, there are several limitations. The severe class imbalance remained a challenge throughout this study despite the efforts to address this using SMOTE and class weighting. Additionally, the data comes from a single academic medical center in the deep south, which limits the generalizability. While the demographics reflect an underrepresented population in broader clinical studies, these results cannot be confidently generalized without external validation. Another limitation was the label construction, where the labels were based on full chart review while the model itself sees only the chief complaints. The model, while certainly much faster than human review, is essentially performing the task of a human in this scenario. Another label limitation was using the fallback method, which could have introduced avoidable noise into the data. Additionally, it is important to note that medical abbreviation expansion was not advised by a clinical professional. Finally, the weighted loss cap of 10 was chosen arbitrarily due to the computational burden tuning that parameter would have placed on an already computationally burdensome study.

Future work should focus on external validation of these results, formally tuning the weighted loss cap (which was set as 10 in this study), consulting a medical professional to inform the medical abbreviation expansion process further, and possibly incorporate additional structured EHR features along with the chief complaint text. In the confusion matrix for the ClinicalBERT test model (**Appendix B**) a few gout cases were missed and had very few of the top 20 words across gout cases, suggesting that these were identified as gout cases upon the supplemental chart review. Supplementing the chief complaints data with EHR would possibly mitigate this.

Conclusion

In conclusion, this study evaluated two feature-based and three representation-learning methods for the task of gout classification from ED chief complaints, identified the best

performing method, and proposed an optimal threshold that could be used in a clinical alert system.

References

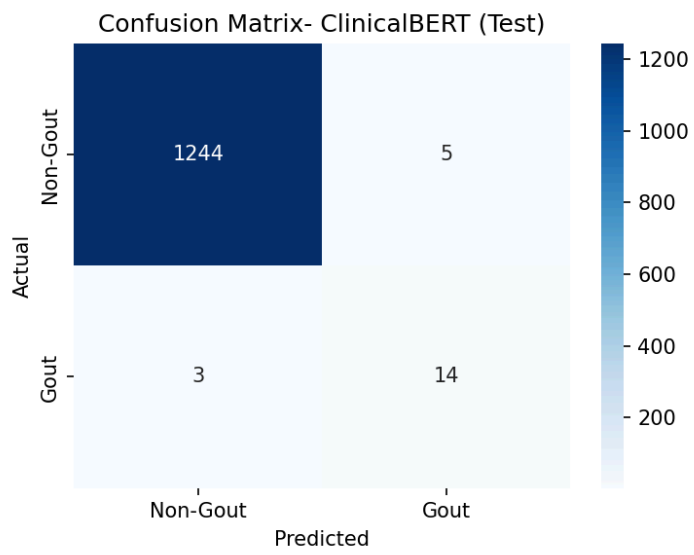
1. American College of Rheumatology. *Understanding Gout: Symptoms & Treatment*. Patient Blog. American College of Rheumatology. <https://rheumatology.org/patient-blog/understanding-gout-symptoms-treatment>
2. Osborne JD, Booth JS, O'Leary T, Mudano A, Rosas G, Foster PJ, Saag KG, Danila MI. Identification of Gout Flares in Chief Complaint Text Using Natural Language Processing. *AMIA Annual Symposium Proceedings*. 2021;2020:973–982. PMID: 33936473. PMC: PMC8075438.
3. Jackson LE, Annapureddy N, Hansen ME, Saag KG, Booth J, Rosas G, Foster PJ, Mudano A, Sun D, Osborne JD, Bongartz T, Hess E, Lawrence C, Dunkel L, Danila MI. Development and Validation of an Emergency Department Electronic Medical Record Gout Flare Alert. *Arthritis Care & Research (Hoboken)*. 2023 Aug;75(8):1821–1829. doi: 10.1002/acr.25061. PMID: 36408730. PMC: PMC10500930.
4. Lopes Oliveira L, Jiang X, Nellippillipathil Babu A, Karajagi P, Daneshkhah A. Effective Natural Language Processing Algorithms for Early Alerts of Gout Flares from Chief Complaints. *Forecasting*. 2024;6(1):224–238. doi: 10.3390/forecast6010013.
5. Osborne, J. D., O'Leary, T., Mudano, A., Booth, J., Rosas, G., Peramsetty, G. S., Knighton, A., Foster, J., Saag, K., & Danila, M. I. (2020). Gout Emergency Department Chief Complaint Corpora (version 1.0). *PhysioNet*. RRID:SCR_007345. <https://doi.org/10.13026/96v3-dw72>
6. Grossman Liu L, Grossman RH, Mitchell EG, Weng C, Natarajan K, Hripcsak G, Vawdrey DK. A deep database of medical abbreviations and acronyms for natural language processing. *Sci Data*. 2021 Jun 2;8(1):149. doi: 10.1038/s41597-021-00929-4. PMID: 34078918; PMCID: PMC8172575.
7. Demner-Fushman D, Chapman WW, McDonald CJ. What can natural language processing do for clinical decision support? *J Biomed Inform*. 2009 Oct;42(5):760-72. doi: 10.1016/j.jbi.2009.08.007. Epub 2009 Aug 13. PMID: 19683066; PMCID: PMC2757540.

Appendix A: Top 20 Words EDA

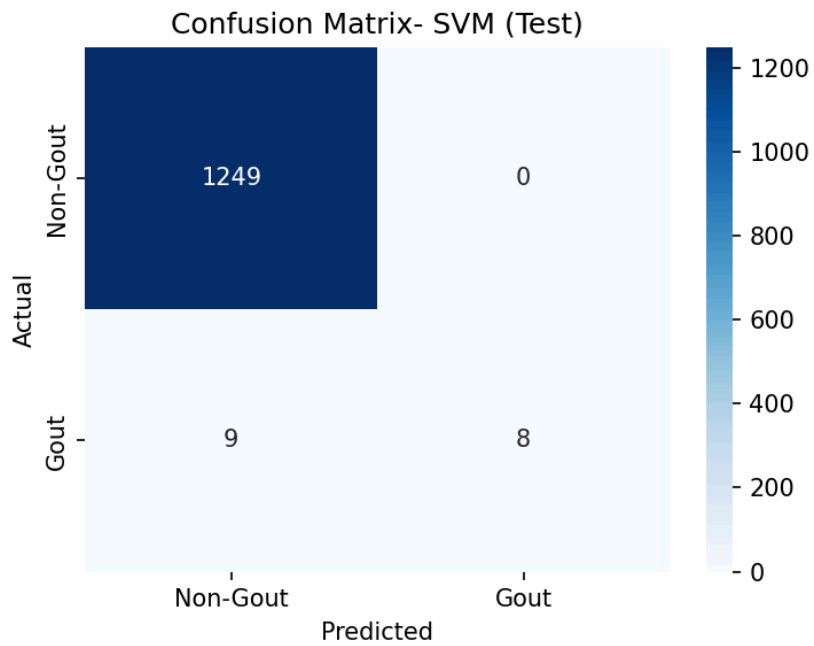
Top 20 words (gout class):

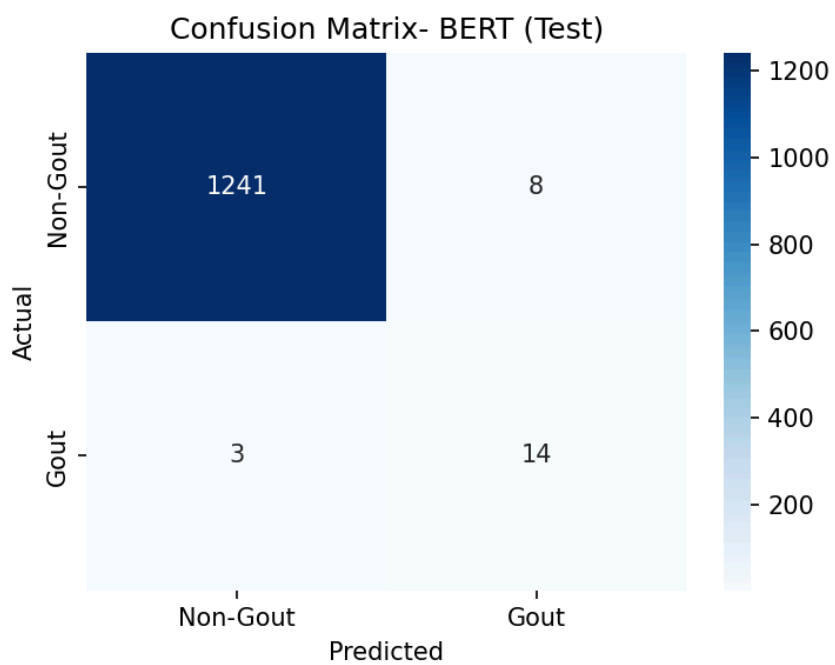
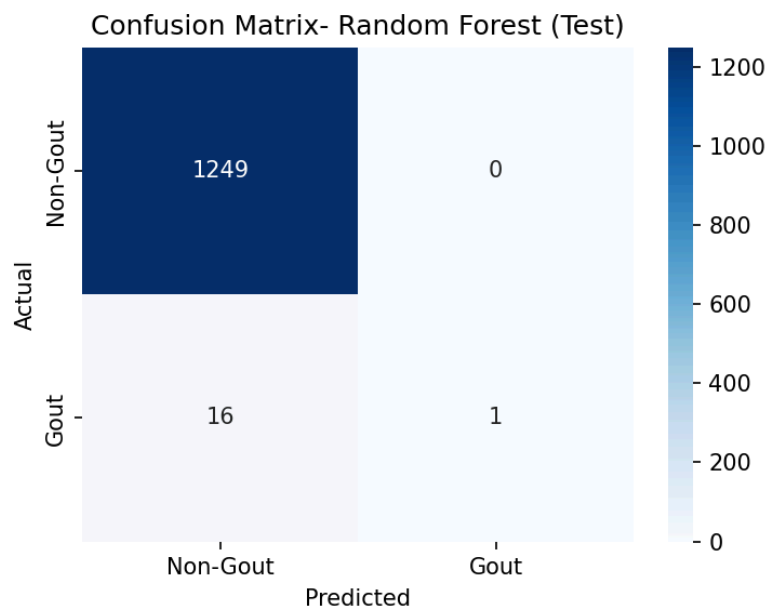
gout: 139
pain: 108
pmh: 58
htn: 45
foot: 36
swelling: 35
denies: 30
knee: 26
left: 26
days: 26
right: 18
pmhx: 18
flare: 16
injury: 15
weeks: 14
states: 14
since: 13
week: 12
ankle: 11
hand: 11

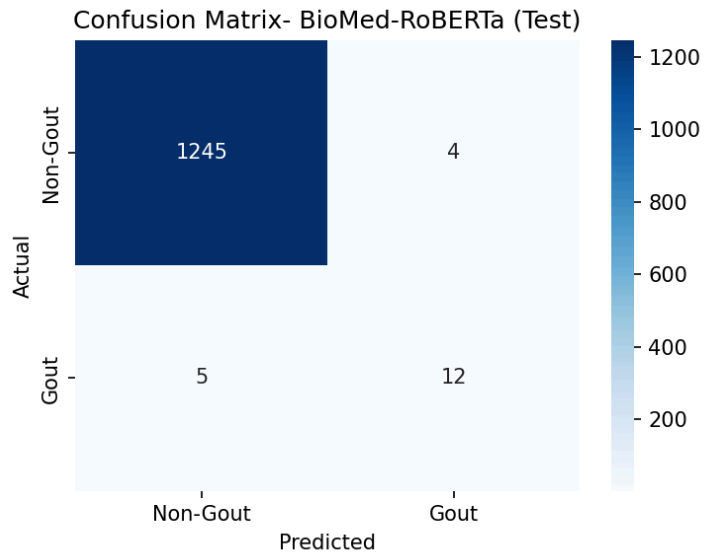
Appendix B: CM for Test ClinicalBERT



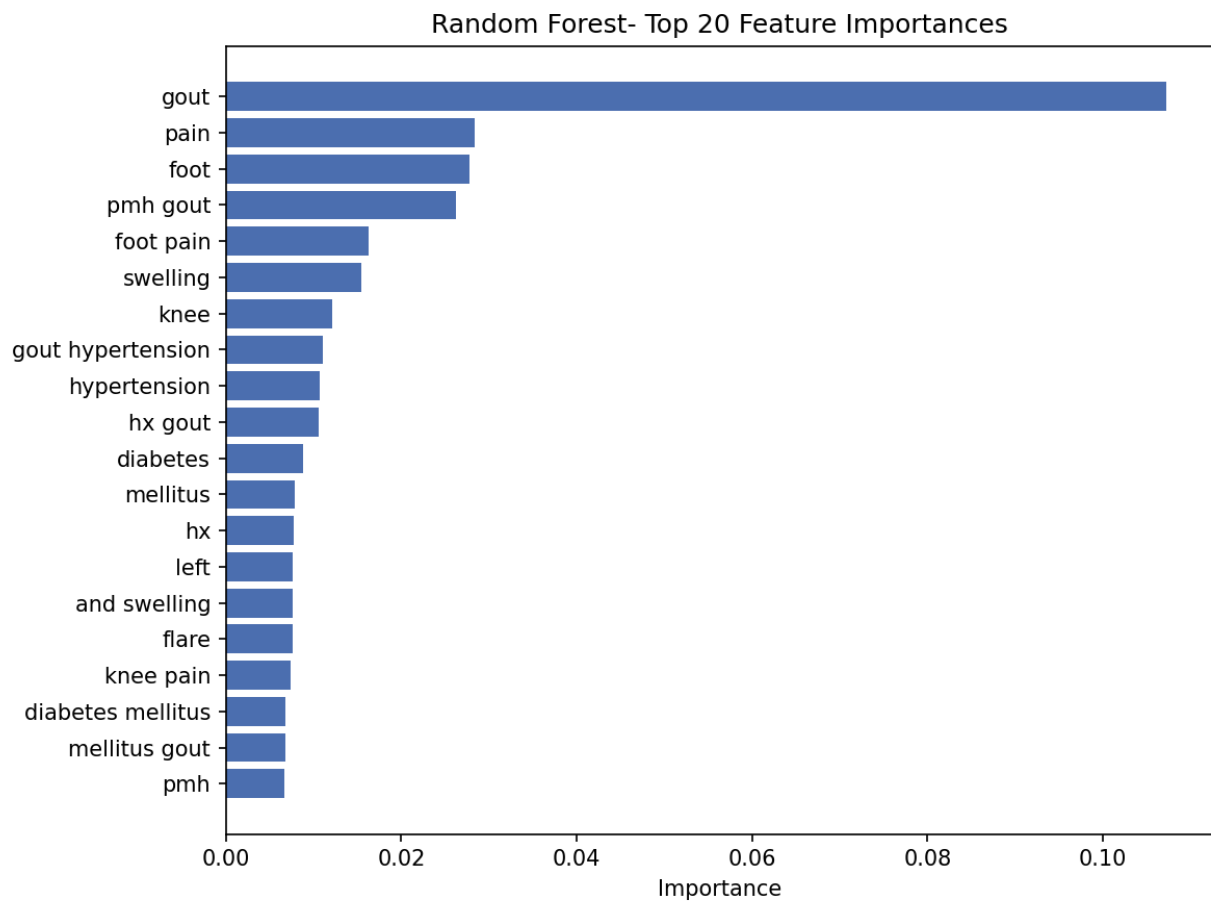
Supplementary FiguresA: Confusion Matrices of all other models







Supplementary Materials B: Random Forest Top Features



Supplementary Materials C: Training Curves (training loss and validation f1)

